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Lesson 1-1

Development of Information Technology and Social Transformation

Learning Objectives

- Explain the major stages in the development of information technology and their impact on society.
- Give examples of social changes and emerging technologies brought about by information technology, and explain their characteristics.

On an ordinary day, a student in Egypt checks messages on an SNS app, pays for breakfast with a cashless app, joins a lesson through online learning, and orders a book from an e-commerce (EC) shop. Twenty years ago, most of this was not possible. Information technology has developed through a series of stages — from the first computers to cloud computing — and at each stage it has changed the way people communicate, work, learn, and pay.

? Guiding question: How has information technology developed through its major stages, and how has each stage changed society?

Explore (in pairs)

With a partner, look at the five time periods in the table below before you read on. For each period, name one device or service from that era that you still use or hear about today. Then predict which stage changed daily life the most, and give one reason.

Point!

1. The History of Information Technology (IT)

(1) The major stages in the development of information technology are summarized in the table below.

Time period	Major technologies and events	Impact on society
1940s-60s	Birth of the computer (ENIAC, vacuum tubes)	Mainly used for military and scientific computation
1970s-80s	Spread of personal computers (PCs)	Beginning of personal computer use
1990s	Commercialization of the Internet; the Web	Globalization of information; spread of email
2000s	Rise of smartphones (iPhone, etc.)	Explosive spread of mobile Internet
2010s onward	Spread of cloud computing	Large-scale data analysis and AI; "IT as a service" becomes widespread

(2) **(Moore's Law)**...the empirical observation that "the number of transistors on an integrated circuit doubles approximately every two years."

Key Fact:

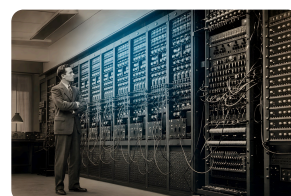
At each stage, information technology introduced a new technology or service and also changed how society communicates, works, and does business.

Key Concepts:

- Moore's Law
- SNS
- e-commerce
- remote work
- online learning
- cashless payment
- edge computing
- autonomous driving
- AR / VR
- quantum computing



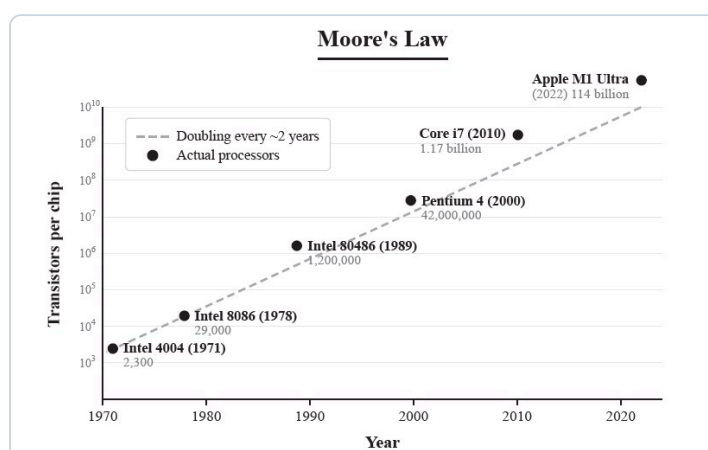
IT advanced in stages, each smaller and more connected.



The first computers filled a whole room.

This remained largely accurate for many years, and computing power increased dramatically.

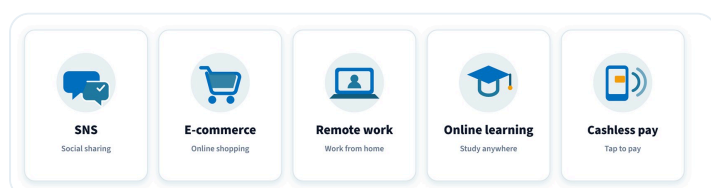
However, in recent years, miniaturization of transistors has been approaching a physical limit. If circuits are made any smaller, problems arise such as the quantum tunneling effect, in which electrons slip through barriers, and leakage current, in which current escapes unintentionally, making it difficult to achieve both higher performance and lower power consumption at the same time. In response, new directions for performance improvement are being pursued, such as parallel processing using multiple processor cores, and quantum computers based on the principles of quantum mechanics.



Moore's Law: transistor count over time

2. Social Changes Resulting from Information Technology

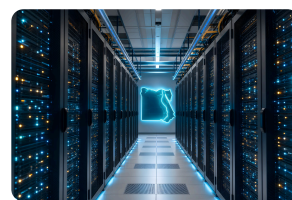
- (1) **(SNS (Social Networking Service))**...services that allow users to connect with each other and post and share information. They are highly effective at spreading information rapidly.
- (2) **(E-commerce (EC))**...buying and selling goods and services through the Internet.
e.g., online shops such as Amazon and eBay
- (3) **(Remote work)**...a working style in which work is performed from home or other remote locations using the Internet.
- (4) **(Online learning)**...a learning style in which classes and study materials are delivered using the Internet.
- (5) **(Cashless payment)**...a system for making payments using electronic money, QR codes, etc., without using cash.
e.g., credit cards, debit cards, mobile payment apps



Five social changes driven by information technology.

KEY TERMS

Moore's Law — the observation that the number of transistors on a chip roughly doubles about every two years.



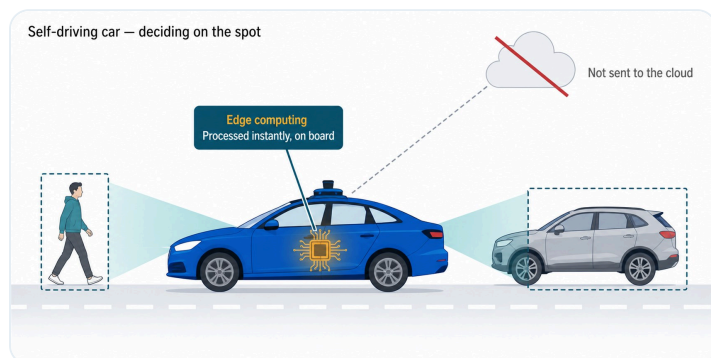
This is "the cloud" — real machines in a real place.

PAUSE & THINK

Of these five changes, which would be hardest to give up — and why?

3. Notable Emerging Technologies

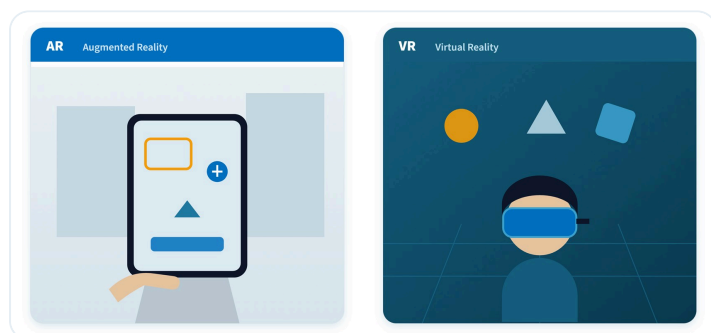
(1) **(Autonomous driving)**...a technology that uses AI to drive a vehicle without human operation. It uses cameras and sensors to recognize the surroundings, makes driving decisions, and controls a vehicle. Because a delay of even 0.1 seconds can lead to an accident, edge computing, in which processing is performed instantly on the vehicle itself rather than sending data to the cloud for judgment, is used.



A self-driving car decides on board (edge computing), without waiting for the cloud.

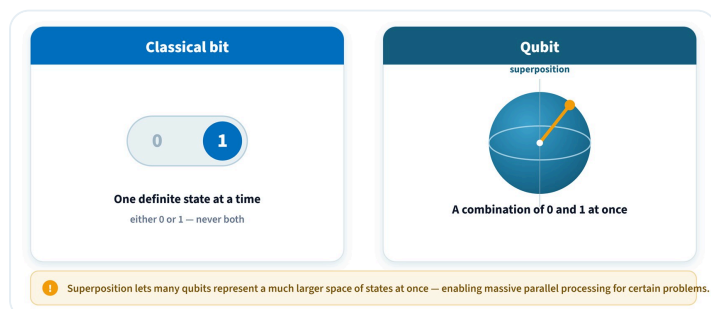
(2) AR / VR

- ① **(AR (Augmented Reality))**...a technology that overlays digital information on real-world images.
- ② **(VR (Virtual Reality))**...a technology that allows users to immerse themselves in a virtual space generated by a computer.



AR adds digital layers to the real world; VR replaces it with a virtual one.

(3) **(Quantum computing)**...a technology expected to dramatically speed up computations that are difficult or impossible for traditional computers, by using the principles of quantum mechanics.



A classical bit holds one state; a qubit uses superposition.

KEY TERMS

edge computing — processing data on the device itself, instantly, instead of sending it to the cloud.

PAUSE & THINK

In autonomous driving, why is it necessary to process data instantly on the vehicle side using edge computing, rather than sending the data to the cloud for judgment? Explain your answer.

✎ Worked Example

(1) From the following options (A–D), choose the one that lists the stages of information technology (IT) development in the correct chronological order.

- A** Birth of the computer → Rise of smartphones → Commercialization of the Internet → Spread of cloud computing
- B** Birth of the computer → Commercialization of the Internet → Rise of smartphones → Spread of cloud computing
- C** Commercialization of the Internet → Birth of the computer → Spread of cloud computing → Rise of smartphones
- D** Rise of smartphones → Commercialization of the Internet → Birth of the computer → Spread of cloud computing

(2) Mark each statement (A–D) with ○ if true or × if false.

- A** Moore's Law is the empirical observation that "the number of transistors on an integrated circuit doubles approximately every two years."
- B** Moore's Law has been said to be approaching a physical limit in recent years.
- C** SNS is highly effective at spreading information rapidly.
- D** E-commerce (EC) refers to purchasing goods at physical stores using cash.

(3) Match each description (a–c) with the most appropriate technology from the choices below (A–C).

- a** A technology that overlays digital information on real-world images
- b** A technology that uses AI to drive a vehicle without human operation
- c** A technology that allows users to immerse themselves in a virtual space generated by a computer

[Choices] **A** Autonomous driving **B** AR **C** VR

KEY TERMS**Moore's Law** —

transistors per chip roughly double every two years.

edge computing — data processed on the device itself, instantly.

✓ Solution

(1) The correct order is "Birth of the computer (1940s-60s) → Commercialization of the Internet (1990s) → Rise of smartphones (2000s) → Spread of cloud computing (2010s onward)." Therefore, **B**.

(2)

- A** Correct definition of Moore's Law. Therefore, ○.
- B** It has been said to be approaching a physical limit in recent years. Therefore, ○.
- C** SNS is a service highly effective at spreading information rapidly. Therefore, ○.
- D** E-commerce refers to buying and selling through the Internet, not purchases at physical stores using cash. Therefore, ×.

(3)

- a:** Overlaying digital information on real-world images is AR (Augmented Reality). **B**
- b:** Driving a vehicle without human operation is autonomous driving. **A**
- c:** Immersing oneself in a virtual space is VR (Virtual Reality). **C**

Try

1 Answer the following questions.

- What is the name of the empirical observation that "the number of transistors on an integrated circuit doubles approximately every two years"?
- What is the term for the buying and selling of goods and services using the Internet?
- What is the term for the working style in which one works from home or other remote locations using the Internet?
- What is the term for the system for making payments using electronic money, QR codes, etc., without using cash?
- What is the term for the technology that uses AI to drive a vehicle without human operation?

2 Answer the following questions.

(1) For each of a–d, indicate which of the following is most closely related: **A** Changes in daily life, **B** Changes in industry and the economy, **C** Changes in healthcare and education.

- a** Taking classes through online learning.
- b** Paying for purchases using a mobile payment app on a smartphone.
- c** Sharing photos with friends on SNS.
- d** A company introducing a work-from-home system.

(2) From the following options (A–D), choose the one that is NOT an appropriate description of an emerging technology.

- A** Autonomous driving uses AI to drive a vehicle without human operation.
- B** AR is a technology that overlays digital information on real-world images.
- C** VR is a technology that dramatically improves the processing speed of a computer.
- D** Quantum computing is expected to speed up computations that are difficult for traditional computers.

EXAM-STYLE QUESTION

Analyze how the spread of cloud computing (from the 2010s onward) has changed the way information technology is used. **[6]** In your answer, refer to: large-scale data analysis, AI, and "IT as a service".

KEY TERMS

remote work — working from home or another remote location using the Internet.

online learning — classes and materials delivered over the Internet.

Think as an Engineer — Investigate & Decide

Start from the advantage and concern you named in the Pause & Think note, then investigate further and decide.

1 • Collect data. Survey ten classmates: do they usually pay with cash or with a cashless method (card, mobile app, or QR code)? Record the results in a table and identify the most common response.

2 • Analyze the stakeholders. Using your survey and your own reasoning, note one benefit and one drawback of a fully cashless society for each group below.

Group	Benefit	Drawback
A customer paying		
A small shop owner		
A person with no bank card or smartphone		

3 • Decide. Using your survey and your table, decide: should your community move toward cashless payment? Recommend one step and give two reasons.

Stuck? Start with the customer — one benefit is speed of payment; one drawback is needing a card or a phone.

PAUSE & THINK

Cashless payment is spreading in many countries. If a fully cashless society were realized, choose one advantage and one possible concern, and briefly explain the reason for each.

In a New Context

A village that has never had Internet access is connected to high-speed Internet and cashless payment for the first time. Using the stages and social changes you studied, predict two ways daily life in the village will change, and identify one new problem the village may face.

KEY TERMS

cloud computing — IT delivered as a service over the Internet.

e-commerce — buying and selling through the Internet.

Lesson Question — Answered

The lesson question asked how information technology developed and how each stage changed society. Information technology advanced through a series of stages rather than in a single step, and each stage changed much more than the technology itself. As computing spread, information became global and then mobile, many everyday activities moved online, and information technology grew into a service that supports large-scale data analysis and AI. Each stage also brought social changes in how people communicate, work, learn, and make payments. Overall, the development of information technology is not only about faster machines; at every stage it has reshaped daily life, industry, and the way society handles information.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

Computers were invented in the 1940s and were initially used mainly for military and scientific computation. Later, in the 1970s and 1980s, (a) became widespread, allowing individuals to use computers. In the 1990s, (b) was commercialized, advancing the globalization of information. In the 2000s, (c) emerged, leading to the explosive spread of mobile Internet.

1. Fill in the blanks (a)–(c).
2. What technology, which spread from the 2010s onward, supports large-scale data analysis and the use of AI?
3. According to Moore's Law, about how often does the number of transistors on an integrated circuit double?

3 Answer the following questions.

(1) From the following options (A–F), choose all that are emerging technologies.

A Email **B** Autonomous driving **C** Personal computers **D** AR **E** VR **F** Quantum computing

(2) From the following options (A–D), choose the one that most appropriately describes the characteristics of SNS.

- A** A service that allows users to connect with each other and post and share information, and is highly effective at spreading information rapidly.
- B** A service for buying and selling goods and services through the Internet.
- C** A working style in which one works from home or other remote locations.
- D** A system for making payments using electronic money or QR codes without using cash.

★ REFLECT & CHALLENGE

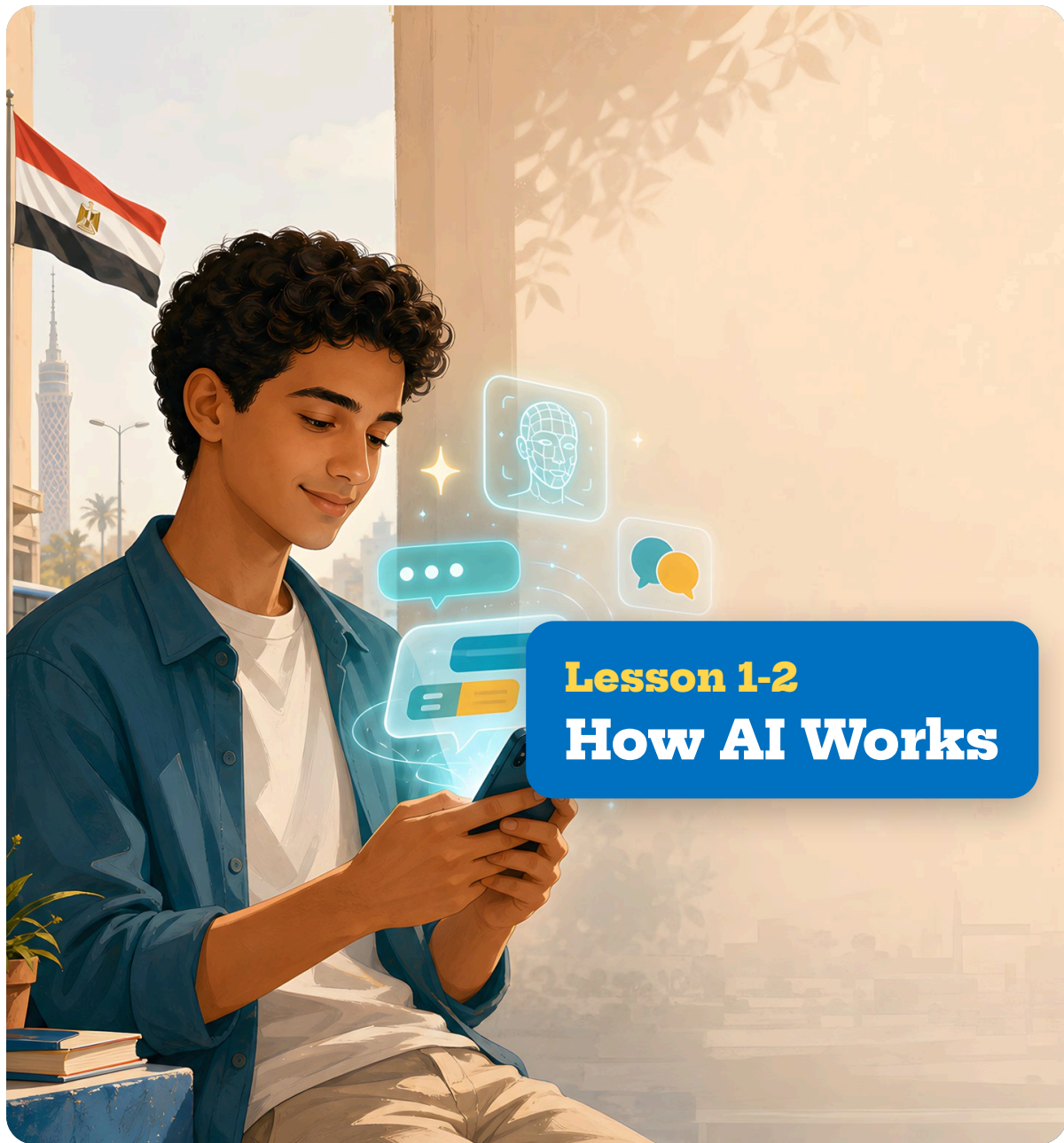
Reflect: Which stage of information technology do you think will matter most in the next ten years? Give one reason. Was your prediction at the start of the lesson correct? What changed your mind? **Challenge:** Choose one emerging technology from this lesson (autonomous driving, AR, VR, or quantum computing). Suggest one way it could help solve a real problem, and state one risk.

KEY TERMS

SNS — connects users to post and share information.
cashless payment — payment without cash (e-money, QR).

★ Key takeaway

Remember: Information technology developed in stages — computers, the Internet, smartphones, and cloud computing. At each stage it introduced a new technology or service and also changed how society communicates, works, learns, and pays.



Learning Objectives

- Explain what AI is.
- Explain how generative AI is positioned within AI technologies.

When you use a phone, AI is often at work. A spam filter sorts your email, a store recommends products you might buy, a translation app changes one language into another, and ChatGPT generates text from a prompt. These are all examples of AI, but they are not all the same kind. AI includes machine learning, within machine learning is deep learning, and today's generative AI systems are built on deep learning — a series of nested technologies, each more specialized than the last.

? Guiding question: What is AI, and how are machine learning, deep learning, and generative AI related?

Explore • In Pairs

With a partner, list three tasks a phone or computer does for you that seem to need "intelligence" (for example, sorting spam, recommending a video, or translating text). Then predict: does the computer follow fixed rules, or does it learn from examples? Give one reason.

Point!

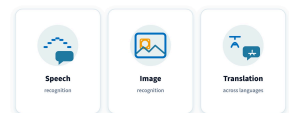
1. What is Artificial Intelligence (AI)?

(1) **(AI (Artificial Intelligence))**...a general term for technologies that reproduce or perform intelligent human behavior (learning, reasoning, judgment, etc.) on a computer.

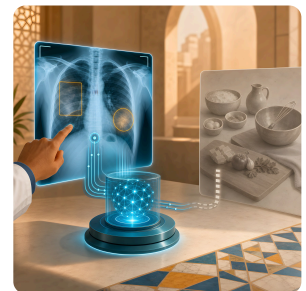
e.g., speech recognition, image recognition, translation

2. The Relationship between AI and Machine Learning

(1) Within AI technologies, there is machine learning; within machine learning, there is deep learning; and within deep learning, there is generative AI. The broad field of AI includes increasingly specialized technologies as a series of nested categories.



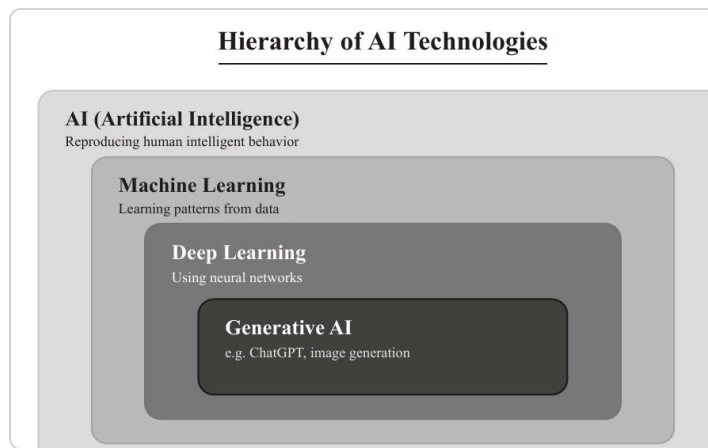
Three everyday AI examples: speech, image, translation.



Today's AI is narrow — expert at one task only.

THINK IT THROUGH

Spam filters and product recommendations are both examples of machine learning, but the tasks they perform are completely different. From the perspective of how AI works, explain what these two applications have in common.



Hierarchy of AI Technologies

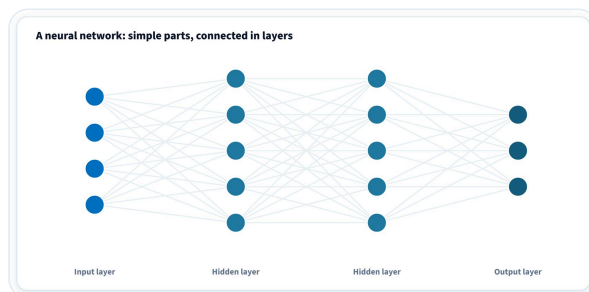
(2) **(Machine Learning)**...one of the learning technologies that makes AI work. It learns patterns from data to make predictions and judgments.

e.g., spam filters, product recommendations

(3) **(Deep Learning)**...an advanced technology within machine learning that uses neural networks. It learns complex patterns using large-scale data.

e.g., image analysis for autonomous driving, speech synthesis

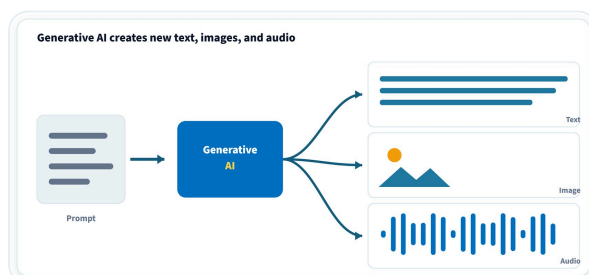
(4) **(Neural network)**...a system modeled after the workings of the nerve cells (neurons) of the human brain. By connecting many components, it learns from data and becomes capable of making complex judgments. It is the core technology supporting recent advances in AI.



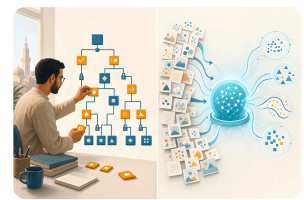
A neural network connects many simple components.

(5) **(Generative AI)**...AI technology that uses deep learning to generate new data (text, images, audio, programs, etc.).

e.g., ChatGPT, image generation AIs



Generative AI creates new text, images, and audio.



Fixed hand-written rules versus rules learned from data.

PAUSE & THINK

Deep learning needs large-scale data to learn. Why might an AI struggle with something it has rarely seen in its data?

KEY TERMS

neural network — a system modeled after the nerve cells (neurons) of the human brain, used to learn from data and make complex judgments.

KEY TERMS

generative AI — AI that uses deep learning to generate new data such as text, images, and audio.

✍ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

- A** Machine learning is one of the technologies that makes AI work.
- B** Deep learning is a completely different technology from machine learning.
- C** Generative AI is a technology that uses deep learning to generate new data.
- D** AI and machine learning have the same meaning.

(2) Match each description (a–c) with the most appropriate term from the choices below (A–C).

- a** A technology that learns patterns from data to make predictions and judgments
- b** A technology that uses neural networks to learn complex patterns
- c** AI technology that generates new data such as text and images

[Choices] **A** Machine learning **B** Deep learning **C** Generative AI

KEY TERMS

machine learning — learns patterns from data.

deep learning — uses neural networks on large-scale data.

✓ Solution

(1)

- A** Machine learning is one of the learning technologies that makes AI work. Therefore, ○.
- B** Deep learning is a type of machine learning (an advanced technology within machine learning). Therefore, ×.
- C** Correct description of generative AI. Therefore, ○.
- D** AI is a broad concept, and machine learning is one of the technologies within it. They do not have the same meaning. Therefore, ×.

(2)

- a:** Learning patterns from data is machine learning. **A**
- b:** Using neural networks is deep learning. **B**
- c:** Generating new data is generative AI. **C**

Try

1 Answer the following questions.

1. What is the general term, expressed by a two-letter abbreviation, for technologies that reproduce or perform intelligent human behavior on a computer?
2. What is the term for the learning technology that makes AI work by learning patterns from data to make predictions and judgments?
3. What is the term for the advanced technology within machine learning that uses neural networks?
4. What is the term for the AI technology that uses deep learning to generate new data?

2 Answer the following questions.

(1) From the following options (A–D), choose the one that correctly describes the relationship between AI, machine learning, deep learning, and generative AI.

- A** AI > Deep Learning > Machine Learning > Generative AI
- B** Machine Learning > AI > Generative AI > Deep Learning
- C** AI > Machine Learning > Deep Learning > Generative AI
- D** Generative AI > Deep Learning > Machine Learning > AI

(2) From the following options (A–D), choose the one that is NOT an appropriate example of generative AI.

- A** ChatGPT
- B** Image generation AI
- C** Spam filter
- D** Audio generation AI

Think as an Engineer — Investigate & Decide

Starting from the Think It Through note about hallucination, investigate further and decide.

1 • Investigate. Find one documented example where a generative AI gave an answer that sounded correct but was actually wrong (a "hallucination"). Record its source — a news article, an official report, or a prompt you tested yourself with a screenshot — and note where you found it. Do not rely on an example recalled from memory without a source.

2 • Weigh it up. Note one benefit and one risk of using generative AI to write a school report — for the student, and for the teacher.

3 • Decide. Should generative AI be allowed for homework in your school? Recommend one rule and give two reasons.

EXAM-STYLE QUESTION

Analyze the relationship between AI, machine learning, deep learning, and generative AI. **[6] In your answer:** use the idea of nested categories, and give one example of each.

THINK IT THROUGH

Generative AI can produce text that sounds plausible but is factually incorrect (a "hallucination"). Explain why it is dangerous to use such output as the answer to a school report as-is, and state what learners should do instead.

Stuck? One benefit is speed; one risk is that the answer may be factually incorrect, so it must be checked.

In a New Context

A farmer wants to use AI to distinguish healthy crops from diseased ones in photographs. Using what you learned about deep learning and image analysis, explain what the AI would need to learn from, and identify one reason its judgment might be wrong.

? Lesson Question — Answered

The lesson question asked what AI is and how machine learning, deep learning, and generative AI are related. AI is a general term for technologies that reproduce or perform intelligent human behavior on a computer, and machine learning, deep learning, and generative AI are increasingly specialized fields nested within it rather than parallel to it. The key idea from this lesson is that these are not four separate technologies to learn in isolation, but one broad field understood at different levels of specialization.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

The general term for technologies that reproduce or perform intelligent human behavior on a computer is (a). One of the learning technologies that makes (a) work by learning patterns from data is (b). Within (b), an advanced technology that uses neural networks is (c). Furthermore, the technology that uses (c) to generate new data is (d).

1. Fill in the blanks (a)–(d).

KEY TERMS

deep learning — uses neural networks to learn complex patterns.

generative AI — uses deep learning to generate new data.

★ REFLECT & CHALLENGE

Reflect: Before this lesson, did you think AI was one single technology? How has your idea changed? Which of the four — AI, machine learning, deep learning, generative AI — was hardest to tell apart, and why?

Challenge: Generative AI can create text, images, and audio. Suggest one helpful use and one possible misuse, and say how the misuse could be reduced.

3 Answer the following questions.

(1) For each example (a–d), indicate which of the following is most closely related: **A** Machine learning, **B** Deep learning, **C** Generative AI.

- a** Automatically sorting spam emails
- b** Recognizing pedestrians on the road in autonomous driving
- c** Automatically generating new images from text
- d** Recommending products based on a user's purchase history

(2) From the following options (A–D), choose the one that most appropriately describes the relationship between AI, machine learning, deep learning, and generative AI.

- A** Machine learning is a type of AI, and deep learning is a type of machine learning.
- B** AI is a type of machine learning, and generative AI is a type of AI.
- C** Deep learning and machine learning are completely different technologies.
- D** Generative AI is a technology that generates data without using machine learning.

★ Key takeaway**Remember:**

AI is the broad field; machine learning sits inside it, deep learning inside machine learning, and today's generative AI is built on deep learning — from the broadest idea to the most specialized.